

No. of Functional Features Included in the Solution

Date	06 July 2026
Team ID	xxxxxx
Project Name	Smart Lender
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	Credit Officer Login	Allows credit officers and financial analysts to create secure accounts and log in to access the application.	Auth Module	Done	0.5
2	Applicant Details Input Form	A responsive web form where users input gender, marital status, education, employment status, income, loan amount, loan term, credit history, and property area.	Frontend (Flask UI)	Done	0.5
3	ML Prediction Engine	The core backend logic that processes validated applicant inputs through the trained XGBoost model to predict loan approval likelihood.	Backend (Scikit-learn, XGBoost)	Done	1.5
4	Multi-Model Comparison Dashboard	Generates comparative accuracy charts for Decision Tree, Random Forest, KNN, and XGBoost to justify the selected model.	Visualization Module	Done	1.0
5	CSV Batch Upload	Allows financial analysts to upload a CSV file containing multiple applicants' data to get bulk loan approval predictions at once.	API / Backend	In Progress	0.5

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
6	Prediction History & Export	Stores past predictions in the database and allows credit officers to download results as a CSV or PDF report.	Database / UI	In Progress	0.5
7	Admin Dataset Management	An admin-only interface to upload new applicant data, clean it, and retrain the ML model.	Admin Panel	Pending	0.25
8	Automated Email Reports	Sends the prediction result and risk classification directly to the credit officer's registered email.	API Integration	Pending	0.25

Feature Summary

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	4
Core / Must-Have Features	4 (Credit Officer Login, Input Form, ML Engine, Model Comparison Dashboard)
Additional / Nice-to-Have Features	4 (Batch Upload, History, Admin Panel, Email Reports)
Features Tested & Verified	4

Feature Category Breakdown

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Applicant Details Input Form, Multi-Model Comparison Dashboard
2	Backend / Logic	3	ML Prediction Engine, Data Preprocessing & Validation, Admin Dataset Management
3	Database / Storage	1	Prediction History Storage (SQLite)
4	API / Integration	1	CSV Batch Upload, Automated Email Reports

S.No	Category	Features in Category	Example Features
5	Security / Authentication	1	Credit Officer Login (Password Hashing, Session Management)